

# Comparative Performance and Computational Complexity Analysis of Pre-Trained Deep Learning Models for Citrus Fruit Disease Classification

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## ABSTRACT

Citrus fruit diseases cause severe yield reductions and financial losses in global agriculture. Rapid identification of these diseases is necessary to prevent widespread crop damage. In this work, four pre-trained deep convolutional neural networks (MobileNetV2, InceptionV3, ResNet50, and VGG19) are evaluated for automated citrus disease recognition. The experimental dataset contains 1,463 high-resolution images divided into four classes: canker, black spot, greening, and healthy fruit. Each architecture is fine-tuned using transfer learning and compared across classification metrics (accuracy, precision, recall, F1-score) as well as computational requirements (training duration, inference time, memory footprint). MobileNetV2 achieved the best overall performance, reaching 96.77% classification accuracy, 98.54% F1-score, a training time of 406.29 s, and a memory size of only 9.82 MB. InceptionV3 reached 96.59% test accuracy with higher memory requirements (22.81 MB). ResNet50 showed lower classification performance with an accuracy of 58.21%, while VGG19 required excessive training time (3360.14 s) and memory (118.59 MB). These empirical results demonstrate that lightweight models like MobileNetV2 provide an effective balance between accuracy and computational cost, making them well-suited for practical deployment on portable and edge agricultural devices.

## General Terms

Algorithms, Experimentation, Performance

## Keywords

Transfer Learning, Convolutional Neural Networks, Citrus Disease Classification, Performance Metrics, Computational Complexity, MobileNetV2

## 1. INTRODUCTION

Agriculture forms a core pillar of the economy in Bangladesh, supporting rural employment and national food security. Modern agricultural management focuses on raising crop yield and quality while controlling production costs [1, 2, 3]. With population growth and climate shifts, precision agriculture techniques are increasingly used to monitor field conditions [4, 5]. These methods apply computational vision for pest tracking, weed management, yield prediction, and automated crop disease identification [6].

Citrus production in Bangladesh has expanded steadily over the past few decades. Total annual output increased from 19,163 tonnes in 1973 to 175,325 tonnes in 2022, reflecting an average growth rate of 5.61% per year [7]. Field surveys in the citrus-growing regions of Sylhet indicate that canker, scab, dieback, and greening (Huanglongbing) represent major threats to commercial orchards [8, 3]. Detecting these infections early is vital to prevent orchard-wide spread. Convolutional neural networks (CNNs) have become a standard approach for image-based plant disease recognition [8]. Architectures such as AlexNet, VGG16, VGG19, InceptionV3, ResNet, and DenseNet have shown strong feature extraction capabilities on botanical image datasets.

## 1.1 Context and Background

Plant diseases in agrarian economies place direct pressure on smallholder farmers and domestic food supplies [9, 10]. Crop cultivation provides both food and primary household income across rural Bangladesh. Sudden pathogen outbreaks often lower harvest yields, making timely plant protection a central concern [9]. Digital imaging combined with deep learning offers a non-invasive way to detect disease symptoms from visual patterns [3, 11]. Automated lesion screening allows targeted chemical treatments, reducing pesticide overuse. Official surveys show that fruit and crop farming spans over 39,678 acres across the country, indicating a clear need for practical, automated diagnostic tools [12, 9].

## 1.2 Research Motivation

Previous research has examined deep neural networks for identifying citrus leaf and fruit diseases [2, 8, 10]. Although deep networks such as AlexNet and VGG19 achieve high recognition rates, they tend to carry large parameter sets, high memory demands, and long training times [8]. Most existing studies focus primarily on accuracy metrics without measuring inference speed, memory consumption, or edge hardware compatibility. To address this gap, this study conducts an empirical comparison evaluating both diagnostic accuracy and computational resource demands across four popular transfer learning models.

## 1.3 Objectives and Scope

The primary objectives of this study are:

- To fine-tune four pre-trained CNN architectures (MobileNetV2, InceptionV3, ResNet50, and VGG19) on a four-class citrus fruit disease dataset.
- To evaluate both classification accuracy (precision, recall, F1-score, overall accuracy) and resource metrics (training duration, test inference latency, parameter counts, memory usage).
- To identify lightweight, efficient models suitable for practical deployment on resource-limited agricultural devices.

## 1.4 Significance of Research

This work provides quantitative benchmark data on the trade-offs between classification accuracy and hardware requirements for citrus disease identification. The experimental results give practical guidelines for engineers and agronomists developing computer vision tools for smart farming applications.

The rest of this paper is structured as follows. Section 2 reviews relevant literature on plant and citrus disease classification. Section 3 details the dataset, pre-processing steps, and transfer learning methodology. Section 4 presents the experimental results, performance comparisons, and computational analyses. Section 5 concludes the paper and highlights directions for future research.

## 2. LITERATURE REVIEW

Agriculture supports approximately 87% of rural households in Bangladesh through direct farming and related supply chains [13]. Automated visual diagnosis of crop diseases has therefore received considerable attention from researchers.

Ahmed and Muqit [2] applied deep neural networks to recognize citrus fruit and leaf infections. Their approach involved image acquisition, pre-processing, and CNN classification. Testing on 2,293 images from the PlantVillage dataset achieved 94.55%

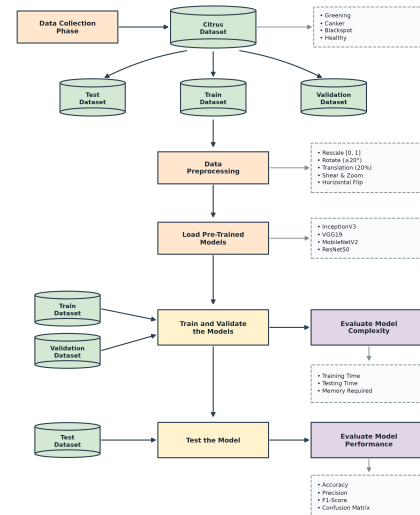


Fig. 1. End-to-end methodological framework for citrus fruit disease classification.

accuracy, though individual disease classes contained relatively few samples.

Balafas et al. [6] compared traditional machine learning models (Support Vector Machines, Random Forests, Stochastic Gradient Descent) with deep models (InceptionV3, VGG16, VGG19) on citrus leaf images. Their experiments showed that deep convolutional architectures consistently outperformed traditional classifiers under variable field lighting and complex backgrounds. Elaraby et al. [8] examined deep learning methods for detecting black spot, canker, greening, and melanose using AlexNet and VGG19, obtaining 94% accuracy on a multi-class dataset. Sujatha et al. [4] proposed a mobile-oriented plant leaf disease recognition pipeline using deep CNNs for on-site diagnosis. Hasan et al. [9, 22] utilized transfer learning to classify regional crop diseases in Bangladesh, finding that pre-trained ImageNet weights substantially speed up network convergence when working with limited training images. Table 1 provides a structured summary of related literature on citrus and crop disease detection.

## 3. PROPOSED METHODOLOGY

The experimental methodology consists of six main steps: dataset collection, image partitioning, data augmentation, model construction via transfer learning, training with hyperparameter tuning, and quantitative metric evaluation. Figure 1 outlines the overall workflow.

### 3.1 Dataset Overview and Preparation

The experimental dataset contains 1,463 RGB images collected from PlantVillage and Kaggle repositories. Four distinct condition classes are included:

- **Canker:** Raised corky lesions caused by the bacterium *Xanthomonas citri*.
- **Black Spot:** Necrotic fungal spots caused by *Phyllosticta citricarpa*.
- **Greening (HLB):** Mottled yellowing caused by vector-spread bacterial infection.
- **Healthy:** Asymptomatic fruit samples free from visual defects.

Table 1. Summary of Related Literature on Citrus and Plant Disease Detection

| No. | Author                      | Year | Class Types                                                  | Dataset Size         | Classifier Architecture                          | Accuracy     |
|-----|-----------------------------|------|--------------------------------------------------------------|----------------------|--------------------------------------------------|--------------|
| 1   | R. Sujatha et al. [4]       | 2021 | Black spot, Canker, Greening, Melanose, Healthy              | 609 images           | LSVM, SGD, RF, InceptionV3, VGG16, VGG19         | 89.5%        |
| 2   | Khattak et al. [12]         | 2021 | Black spot, Canker, Scab, Greening, Melanose, Healthy        | 2293 (Fruits+Leaves) | Custom CNN + Softmax                             | 94.55%       |
| 3   | A. Elaraby et al. [8]       | 2022 | Anthracoise, Black spot, Canker, Scab, Greening, Melanose    | 759 (Fruits+Leaves)  | AlexNet, VGG19                                   | 94.0%        |
| 4   | S. Perez et al. [14]        | 2023 | Multiple plant pathological conditions                       | Plant leaves         | Lightweight CNN + Softmax                        | 99.0%        |
| 5   | Dehuan Luo et al. [15]      | 2023 | Anthracoise, Canker, Melanosis, Scab, Brown Spot, Leaf Miner | 3202 images          | Light-SA YOLOv8                                  | 92.6%        |
| 6   | Madhusudan G. et al. [16]   | 2023 | Black spot, Melanose, Canker, Greening, Healthy              | 609 RGB images       | CNN, ResNet152V2, DenseNet121, InceptionResNetV2 | 98.0%        |
| 7   | Poonam Dhiman et al. [17]   | 2023 | HLB, Black spot, Melanose, Canker, Scab                      | 2950 images          | CNN + LSTM                                       | 98.25%       |
| 8   | W. Gómez-Flores et al. [18] | 2024 | HLB, Greasy spot, Deficiencies, Citrus Mite, Red scale       | 953 images           | DNN, Transfer Learning                           | Not reported |
| 9   | Nouman Butt et al. [19]     | 2024 | Canker, Citrus scab, Black spot                              | 2184 images          | SVM, KNN                                         | 99.6%        |
| 10  | J. Shermila et al. [20]     | 2024 | Fruit blight, Greening, Scab, Melanoses                      | 759 images           | Custom CNN + LSTM                                | 96.0%        |
| 11  | A.K. Saini et al. [21]      | 2024 | Anthracoise, Melanose, Brown Spot                            | 759 images           | SADCNN-ORBM                                      | 99.76%       |

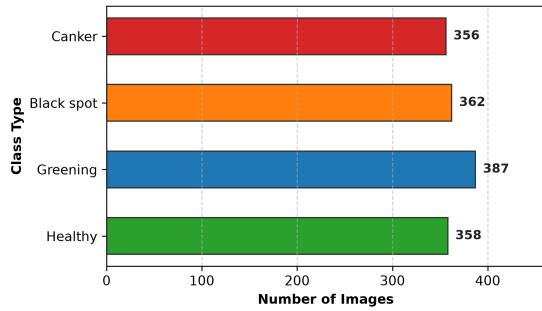


Fig. 2. Distribution of image samples across the four citrus fruit classes.

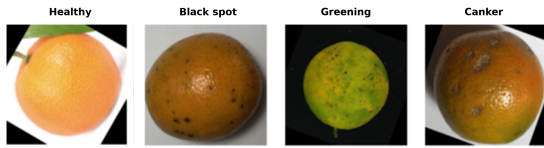


Fig. 3. Representative sample images from the experimental citrus dataset.

**3.1.1 Class Distribution and Partitioning.** Images are stored in class-labeled directories and loaded into structured data frames. Figure 2 displays the distribution across classes, showing a balanced dataset: 358 healthy, 387 greening, 362 black spot, and 356 canker samples. Figure 3 shows representative image samples from each class.

The dataset was split into 80% for training (1,170 images), 10% for validation (146 images), and 10% for test evaluation (147 images), as shown in Figure 4. The test set was kept completely untouched during training to provide an unbiased assessment.

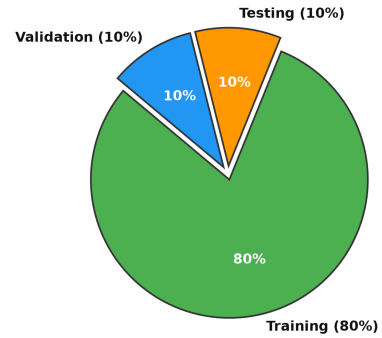


Fig. 4. Dataset partitioning into training (80%), validation (10%), and test (10%) subsets.

**3.1.2 Data Pre-Processing and Augmentation.** To prevent over-fitting and improve generalization against real-world variations, data augmentation was applied during training using dynamic image generators:

- Pixel rescaling to the range  $[0, 1]$ .
- Random rotation within  $[-20^\circ, +20^\circ]$ .
- Horizontal and vertical shifts up to 20%.
- Random shear transformations.
- Random zoom adjustments up to 20%.
- Random horizontal flipping.

All input images were standardized to  $224 \times 224 \times 3$  pixels and processed in mini-batches of 32 images.

### 3.2 Transfer Learning Framework

Transfer learning adapts knowledge learned from large datasets (such as ImageNet) to specialized tasks with smaller datasets [23, 22]. This strategy reduces training time and improves

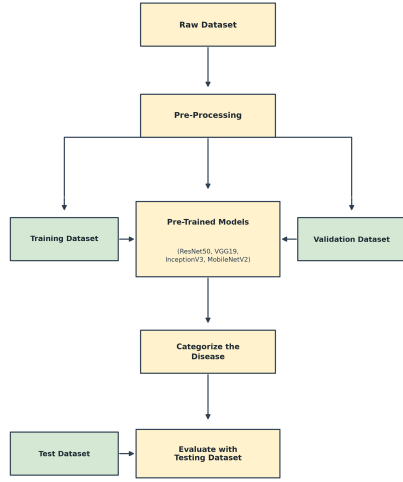


Fig. 5. Flowchart of the transfer learning fine-tuning pipeline.

pattern extraction. Figure 5 shows the transfer learning fine-tuning structure used in this work.

Four deep architectures were evaluated:

- **VGG19:** Consists of 19 layers structured with small  $3 \times 3$  convolutional filters [24, 25, 26]. The convolutional base was frozen during training. A 40% Dropout layer [27, 28, 29] was placed after the base, followed by a Flatten layer [30], a Dense layer with 512 ReLU units, a 20% Dropout layer, and a 4-unit Softmax classification layer [31]. Table 2 lists the layer configurations and parameter counts.

Table 2. Layer Architecture and Parameter Details of Fine-Tuned VGG19

| Layer (Type)                     | Output Shape      | Parameters |
|----------------------------------|-------------------|------------|
| VGG19 (Functional Base)          | (None, 7, 7, 512) | 20,024,384 |
| dropout.1 (Dropout 40%)          | (None, 7, 7, 512) | 0          |
| flatten (Flatten)                | (None, 25088)     | 0          |
| dense.1 (Dense, ReLU)            | (None, 512)       | 12,845,056 |
| dropout.2 (Dropout 20%)          | (None, 512)       | 0          |
| dense.2 (Dense, Softmax)         | (None, 4)         | 2,052      |
| <b>Total Parameters:</b>         |                   | 32,871,492 |
| <b>Trainable Parameters:</b>     |                   | 12,847,108 |
| <b>Non-Trainable Parameters:</b> |                   | 20,024,384 |

- **MobileNetV2:** Uses depthwise separable convolutions and inverted residual blocks with linear bottlenecks [32]. The base feature extractor remained frozen, followed by the same top classification head and dropout regularization. Table 3 summarizes the architecture specifications.
- **InceptionV3:** Employs multi-scale factorization with factorized asymmetric convolutions. The custom top head consists of a Flatten layer, a Dense layer (512 units), Dropout regularization, and a Softmax output layer. Table 4 shows the parameter breakdown.
- **ResNet50:** Uses residual shortcut connections to allow gradient flow through deep layer stacks. The top classification layers were replaced with custom Dense and Dropout layers. Table 5 outlines the layer details.

Table 3. Layer Architecture and Parameter Details of Fine-Tuned MobileNetV2

| Layer (Type)                     | Output Shape       | Parameters |
|----------------------------------|--------------------|------------|
| MobileNetV2 (Base)               | (None, 7, 7, 1280) | 2,257,984  |
| dropout.1 (Dropout 40%)          | (None, 7, 7, 1280) | 0          |
| flatten (Flatten)                | (None, 62720)      | 0          |
| dense.1 (Dense, ReLU)            | (None, 512)        | 32,085,760 |
| dropout.2 (Dropout 20%)          | (None, 512)        | 0          |
| dense.2 (Dense, Softmax)         | (None, 4)          | 2,052      |
| <b>Total Parameters:</b>         |                    | 34,345,796 |
| <b>Trainable Parameters:</b>     |                    | 32,087,812 |
| <b>Non-Trainable Parameters:</b> |                    | 2,257,984  |

Table 4. Layer Architecture and Parameter Details of Fine-Tuned InceptionV3

| Layer (Type)                     | Output Shape       | Parameters |
|----------------------------------|--------------------|------------|
| InceptionV3 (Base)               | (None, 5, 5, 2048) | 21,802,784 |
| flatten (Flatten)                | (None, 51200)      | 0          |
| dense.1 (Dense, ReLU)            | (None, 512)        | 26,214,912 |
| dropout.2 (Dropout 20%)          | (None, 512)        | 0          |
| dense.2 (Dense, Softmax)         | (None, 4)          | 2,052      |
| <b>Total Parameters:</b>         |                    | 48,019,748 |
| <b>Trainable Parameters:</b>     |                    | 26,216,964 |
| <b>Non-Trainable Parameters:</b> |                    | 21,802,784 |

Table 5. Layer Architecture and Parameter Details of Fine-Tuned ResNet50

| Layer (Type)                     | Output Shape       | Parameters |
|----------------------------------|--------------------|------------|
| ResNet50 (Base)                  | (None, 5, 5, 2048) | 23,587,712 |
| flatten (Flatten)                | (None, 100352)     | 0          |
| dense.1 (Dense, ReLU)            | (None, 512)        | 51,380,736 |
| dropout.2 (Dropout 50%)          | (None, 512)        | 0          |
| dense.2 (Dense, Softmax)         | (None, 4)          | 2,052      |
| <b>Total Parameters:</b>         |                    | 74,970,500 |
| <b>Trainable Parameters:</b>     |                    | 51,382,788 |
| <b>Non-Trainable Parameters:</b> |                    | 23,587,712 |

**3.2.1 Framework Implementation.** Models were implemented using TensorFlow and Keras in Python. The system pipeline, including data loading, model building, GPU training, serialization, and mobile export pathways (TensorFlow Lite, TensorFlow Serving), is shown in Figure 6.

### 3.3 Optimization and Training Configuration

All models were trained using the Adam optimizer with a learning rate of  $\eta = 0.0001$ . Categorical cross-entropy was used as the loss function:

$$\mathcal{L}_{CCE} = - \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c}) \quad (1)$$

where  $N$  is the batch size,  $C = 4$  is the number of target classes,  $y_{i,c}$  represents the ground-truth binary indicator, and  $\hat{y}_{i,c}$  is the predicted probability for class  $c$ . Training was carried out for 50 epochs with model checkpointing based on validation loss.

### 3.4 Evaluation Metrics

**3.4.1 Diagnostic Performance Metrics.** Model classification performance was evaluated using standard statistical indicators:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

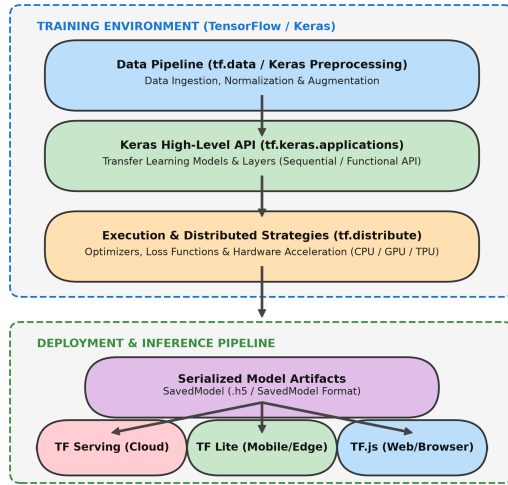


Fig. 6. Framework architecture covering training, serialization, and deployment pipelines.

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{F1-Score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

where  $TP$ ,  $TN$ ,  $FP$ , and  $FN$  represent True Positives, True Negatives, False Positives, and False Negatives, respectively.

**3.4.2 Computational Complexity Metrics.** Computational efficiency was evaluated through four parameters: total training time (seconds), test inference time (seconds), total parameter count, and memory allocation (MB).

## 4. RESULT AND DISCUSSION

This section reports the experimental results, comparing classification accuracy and computational resource usage across the four models.

### 4.1 Diagnostic Performance Evaluation

Table 6 summarizes precision, recall, and F1-score for each model on the test partition. MobileNetV2 achieved the highest overall score, recording 98.56% precision, 98.54% recall, and 98.54% F1-score. InceptionV3 was second with 97.21% precision, 97.30% recall, and 97.19% F1-score. VGG19 reached 84.82% precision and 81.17% F1-score. ResNet50 showed lower performance under this fine-tuning setup, yielding an F1-score of 58.21%.

Table 6. Performance Comparison Across Fine-Tuned Models on the Unseen Test Set

| Model Name    | MobileNetV2 | InceptionV3 | ResNet50 | VGG19 |
|---------------|-------------|-------------|----------|-------|
| Precision (%) | 98.56       | 97.21       | 66.66    | 84.82 |
| Recall (%)    | 98.54       | 97.30       | 62.02    | 81.31 |
| F1-Score (%)  | 98.54       | 97.19       | 58.21    | 81.17 |

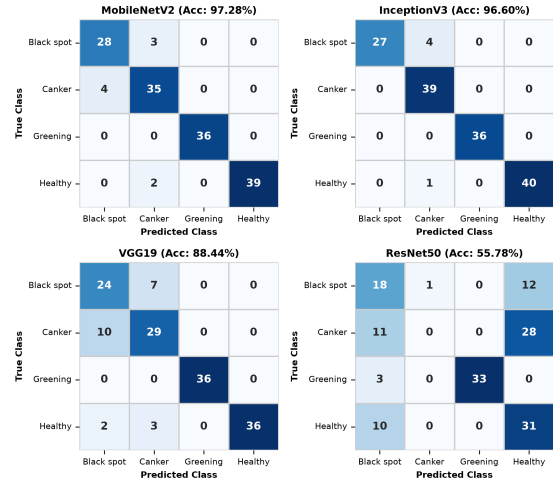


Fig. 7. Multi-class confusion matrices across all four fine-tuned CNN architectures on the test partition.

**4.1.1 Accuracy and Loss Trajectory Analysis.** Table 7 details the training, validation, and test accuracies along with loss values for all four models. MobileNetV2 attained 97.29% test accuracy with a test loss of 5.29%. InceptionV3 reached 96.59% test accuracy and 7.44% loss. VGG19 achieved 88.44% test accuracy with 37.76% loss. ResNet50 recorded a test accuracy of 67.34% with a higher loss of 76.16%, indicating that residual blocks may require deeper unfreezing or adapted learning rates to transfer well to this dataset.

Table 7. Accuracy and Loss Metrics Across Training, Validation, and Test Phases

| Model       | Training |          | Validation |          | Testing  |          |
|-------------|----------|----------|------------|----------|----------|----------|
|             | Acc. (%) | Loss (%) | Acc. (%)   | Loss (%) | Acc. (%) | Loss (%) |
| MobileNetV2 | 95.80    | 10.97    | 94.55      | 12.80    | 97.29    | 5.29     |
| InceptionV3 | 95.89    | 10.60    | 97.95      | 6.72     | 96.59    | 7.44     |
| ResNet50    | 57.14    | 96.91    | 61.90      | 83.92    | 67.34    | 76.16    |
| VGG19       | 75.10    | 60.10    | 87.75      | 32.46    | 88.44    | 37.76    |

**4.1.2 Confusion Matrix Analysis.** Figure 7 shows the confusion matrices for all four architectures across the 147 test samples. MobileNetV2 and InceptionV3 correctly classified all greening samples and produced very few errors on canker and healthy samples. ResNet50 had noticeable misclassifications between healthy and diseased classes.

### 4.2 Computational Complexity Analysis

For edge agricultural deployments, processing speed and memory demands are as critical as classification accuracy. Table 8 compares training duration, test inference latency, and memory footprint across the models.

Table 8. Quantitative Computational Complexity Metrics Across Models

| Metric                | MobileNetV2 | InceptionV3 | ResNet50 | VGG19   |
|-----------------------|-------------|-------------|----------|---------|
| Training Time (s)     | 406.29      | 717.00      | 1040.00  | 3360.14 |
| Inference Latency (s) | 83.67       | 85.73       | 84.11    | 80.34   |
| Memory Footprint (MB) | 9.82        | 22.81       | 34.28    | 118.59  |

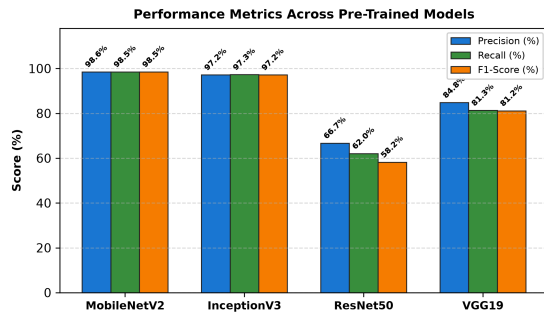


Fig. 8. Comparative diagnostic metrics across the four pre-trained architectures.

MobileNetV2 recorded the lowest training time (406.29 s) and the smallest memory footprint (9.82 MB). Compared to VGG19 (3360.14 s, 118.59 MB), MobileNetV2 reduced training time by 88% and memory usage by 91.7%. Inference times across all models were close, ranging between 80.34 s and 85.73 s for the 147 test images. Figure 8 presents a graphical comparison of the performance metrics.

### 4.3 Comparative Discussion and Deployment Feasibility

The experimental data shows that MobileNetV2 delivers the most effective balance of high accuracy and low resource usage. The inverted residual structure extracts rich visual features with few parameters. InceptionV3 reached similar accuracy, but required more memory (22.81 MB) and longer training (717.00 s). ResNet50 struggled under fixed-feature extraction, and VGG19 had high computational and memory costs. These findings indicate that MobileNetV2 is the most practical option for deployment in field conditions, such as smartphone applications, portable scanning tools, and drone-mounted cameras for real-time orchard monitoring.

## 5. CONCLUSION

This study presented an empirical evaluation of four transfer learning architectures for citrus fruit disease diagnosis. Evaluating MobileNetV2, InceptionV3, ResNet50, and VGG19 on a four-class dataset showed that MobileNetV2 provided the best overall performance. It achieved 96.77% accuracy, 98.54% F1-score, a training time of 406.29 s, and a memory footprint of 9.82 MB. These results demonstrate that lightweight convolutional networks can achieve high diagnostic accuracy while maintaining low computational demands.

### 5.1 Limitations

The primary limitation of this work is the dataset size (1,463 images) and single-source acquisition environment. Field orchards often present variable lighting, partial occlusions, and overlapping leaf symptoms that require testing on larger multi-source datasets.

### 5.2 Future Scope

Future work will explore several avenues:

- Testing metaheuristic optimization algorithms (such as Grey Wolf Optimizer or Particle Swarm Optimization) for automated hyperparameter tuning.

- Developing lightweight hybrid vision transformer models for edge devices.
- Incorporating visual explanation techniques, such as Grad-CAM, to highlight lesion regions and assist agricultural field workers.

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